

Mastering Spring AI: Building Production-Ready AI Applications

Comprehensive Course Syllabus & Enterprise Blueprint for Java Developers

Module 1: Introduction to Spring AI & Ecosystem

Establish a solid foundation. Learn the core philosophy behind Spring AI, understand the unified API abstraction layer, and discover how it seamlessly integrates with standard Spring Boot principles. Set up your development environment and configure API authentication natively using standard properties configurations.

Module 2: Text Generation & Advanced Prompt Engineering

Dive deep into ChatClient and ChatModel abstractions. Master dynamic template configurations, system prompt isolations, and advanced contextual boundaries. Learn to handle structured outputs using ChatOptions, mapping raw LLM responses directly into custom Java POJOs cleanly.

Module 3: Embedding Models & Semantic Vector Spaces

Understand the mathematics and architecture of dense vector spaces. Implement text tokenization workflows using TokenTextSplitter builders. Learn how modern models like gemini-embedding-2 compress textual context, and explore Matryoshka compression mechanics to tailor dimensionality dynamically to application constraints.

Module 4: Vector Databases & High-Performance Storage

Integrate enterprise-grade vector stores natively using pgvector and PostgreSQL. Master advanced index patterns including FLAT and HNSW (Hierarchical Navigable Small World) to manage highly dimensional spaces efficiently. Write sophisticated metadata filter expressions to narrow search scopes rapidly.

Module 5: Implementing RAG (Retrieval-Augmented Generation)

Build an end-to-end production RAG pipeline. Extract content efficiently from complex multi-page PDF documents using TikaDocumentReader. Implement automated text chunking strategies with optimal token overlaps, populate the vector store dynamically in controlled batches, and leverage context injection to defeat LLM hallucinations.

Module 6: State Management, Chat Memory & Advisors

Master conversational state management in Spring AI. Dive deep into the ChatClient Advisor architecture. Implement MessageChatMemoryAdvisor to provide LLMs with conversational context seamlessly, and configure production-ready state retention using JdbcChatMemory to persist user sessions across server restarts.

Module 7: Extending Capabilities with MCP (Model Context Protocol) & Tool Calling

Bridge the gap between AI intuition and enterprise infrastructure using the revolutionary Model Context Protocol (MCP). Connect your Spring Boot app to external MCP servers to fetch live enterprise data, read local file structures, and register standard Java beans as tools using automated Function Calling frameworks.

Module 8: Enterprise Observability & Monitoring

Bring transparency to your production AI workflows. Integrate Micrometer Observation and Spring Boot Actuator to trace LLM latency, token usage, and embedding performance. Export structured metrics and logs to platforms like Prometheus and Grafana to monitor cost, accuracy, and system health in real time.